

10 CHEBYSHEV AND THE HIDDEN LAYER (lectures 10-11 + H1-H8)

PART A: CHEBYSHEV'S INEQUALITY

1. The two forms, and why they exist

The theorem slide states BOTH:

$$\begin{aligned} \square P(|X - \mu| \geq k) &\leq \sigma^2/k^2 && \text{(the definition, } k \text{ in the EVENT)} \\ \square P(|X - \mu| \geq k \cdot \sigma) &\leq 1/k^2 && \text{(the } k\text{-sigma restatement)} \end{aligned}$$

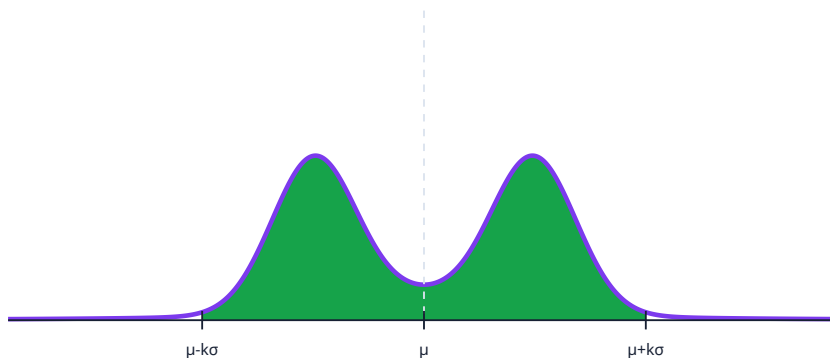
and the complement form:

$$\square P(|X - \mu| < k \cdot \sigma) \geq 1 - 1/k^2 \quad \text{(for "at least"/"within")}$$

CHEBYSHEV NEEDS ONLY μ AND σ . NO DISTRIBUTION IS ASSUMED.
It is always an UPPER BOUND on a TAIL.

2. The geometry, drawn

Chebyshev: guaranteed mass around the mean



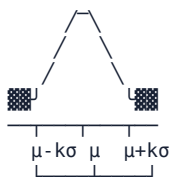
$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$$

valid for any distribution with finite mean and variance

schematic density, not a normal-distribution assumption

Chebyshev central-mass guarantee without a shape assumption

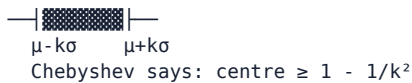
the event $|X - \mu| \geq k\sigma$ is the union of TWO tails:



each tail is part of the event
Chebyshev says: total tail $\leq 1/k^2$

and the complement $|X - \mu| < k\sigma$ is the CENTRE:

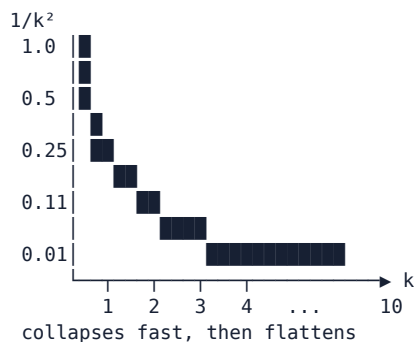




3. The thresholds table (memorise these)

k	$1/k^2$ tail	$1 - 1/k^2$ centre
1	1.0000	0.0000 (vacuous)
2	0.2500	0.7500 = 3/4
3	0.1111	0.8889 = 8/9
4	0.0625	0.9375 = 15/16
5	0.0400	0.9600 = 24/25
10	0.0100	0.9900 = 99/100

the shape of the bound:



4. The five facts the paper probes (formula sheet section E)

1. only a finite mean and variance are needed, NO distribution assumed
2. $k \leq 1$ gives a vacuous bound (says only " ≤ 1 ")
3. the bound is an UPPER bound on the tail, so a known-distribution question wants BOTH the bound and the exact value
4. k is measured in STANDARD DEVIATIONS, so convert $k = \varepsilon/\sigma$ first
5. the exact useful values: $k=2 \rightarrow 3/4$, $k=3 \rightarrow 8/9$, $k=4 \rightarrow 15/16$

5. WORKED: the corpus questions

— Q1 (safe) —

X has $E(X) = 3$ and $E(X^2) = 13$. Find a lower bound for $P(-2 < X < 8)$.

$$\text{Var} = E(X^2) - [E(X)]^2 = 13 - 9 = 4, \quad \sigma = 2$$

$$-2 < X < 8 \text{ is } |X-3| < 5, \text{ so } k = 5/2 = 2.5$$

$$P(|X-3| \geq 5) \leq 4/25$$

$$\text{complement } P(-2 < X < 8) = P(|X-3| < 5) \geq 21/25$$

— Q2 (safe) —

$\mu = 10$, $\sigma^2 = 4$ (so $\sigma = 2$)

$$P(|X - 10| \geq 3): \quad k = 3/2 = 1.5$$

$$\text{bound} = 1/k^2 = 1/2.25 = 0.4444 = 4/9$$

$$\text{complement } P(|X-10| < 3) \geq 1 - 4/9 = 5/9 = 0.5556$$

and the C=10 form: find c so $P(|X-10| \geq c) \leq 0.04$

$$1/k^2 = 0.04 \rightarrow k^2 = 25 \rightarrow k = 5$$

$$c = k \cdot \sigma = 5 \times 2 = 10$$

ERRATA 15: the deck's Q3 slide is a MESS.
 - it STATES the row $(-1, -1, 3, 5)$ with $p = 1/6, 1/6, 1/6, 1/2$
 - it WORKS the row $(-1, +1, 3, 5)$ (the second value +1)
 - it writes a reciprocal template $P(|x-\mu| \geq k) < k^2/\sigma^2$

USE Q1 AND Q2 ONLY. Do not quote Q3's row.

(For the record, the WORKED row gives $E=3$, $E(X^2)=43/3$,
 $\text{Var}=16/3$, bound $16/3$. The STATED row would give $E=8/3$,

Var=65/9. They are different distributions.)

— WORKED: two dice (A1 short Q5) —
 Var(one die) = 35/12
 Var(sum of two) = 35/12 + 35/12 = 35/6 <- INDEPENDENCE RULE

$P(|X-7| \geq 3)$: $k = 3$, $\sigma^2 = 35/6$
 bound = $1/k^2 = 1/9$... no wait, use form $\square$:
 bound = $\sigma^2/k^2 = (35/6)/9 = 35/54 = 0.6481$

exact: sums ≤ 4 or ≥ 10 are 12 of 36 outcomes $\rightarrow 1/3 = 0.3333$

bound holds ($0.648 \geq 0.333$) and is much wider, as expected

— WORKED: 600 throws (A1 short Q6) —
 $X \sim B(600, 1/6)$: $\mu = 100$, $\sigma^2 = npq = 600(1/6)(5/6) = 250/3$

$P(80 \leq X \leq 120) = P(|X-100| \leq 20)$
 $\epsilon = 20$, $k^2 = \epsilon^2/\sigma^2 = 400/(250/3) = 1200/250 = 4.8$
 Chebyshev directly bounds the strict complement $|X-100| > 20$
 by at most $5/24$, so the inclusive event is at least $19/24$.

exact binomial = 0.9754

— WORKED: marks (A1 app Q4) —
 $\mu = 70$, variance = 25 ($\sigma = 5$). $P(60 < X < 80)$?
 $|X-70| < 10$, $k = 10/5 = 2$
 $P \geq 1 - 1/4 = 0.75$

6. The k-extraction recipe

1. get μ and σ ($\sigma = \sqrt{\text{Var}}$, compute Var if not given)
2. write the event as $|X - \mu| \geq \epsilon$ or $< \epsilon$
3. $k = \epsilon / \sigma$ <- RAW UNITS TO SIGMA UNITS
4. tail form $1/k^2$, or complement $1 - 1/k^2$

PART B: THE HIDDEN LAYER (H1-H8)

These carry marks and have NO teaching slide. Full treatment in [../11-BEYOND-SLIDES.md](#).

H1 INDEPENDENCE RULES OF RVs
 $E(XY) = E(X)E(Y)$ [needs independence]
 $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ [needs independence]
 $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ [the minus STAYS PLUS]
 $E(X+Y) = E(X) + E(Y)$ [ALWAYS, no condition]
 gates: dice-sum variance, $\text{Var}(X^-)$, estimator variances,
 Poisson additivity, every CLT SE

H2 CHEBYSHEV (the whole topic) - taught only in the L10-11
 deck, which arrived 15 Sep. Graded 9+ times.

H3 MEMORYLESS $P(T>s+t | T>s) = P(T>t)$
 exponential never ages. $P(T>t) = e^{(-\lambda t)}$.
 graded: A2 MCQ Q5, A2 section C Q1.

H4 NORMAL LANDMARKS 68.27 / 95.45 / 99.73 within 1,2,3 sigma
 graded: A2 MCQ Q11.

H5 EXPECTED COUNT = $N \times P$
 "out of N items, how many do we expect"

graded: A1 app, A2 B4 (5000 batteries), A2 D1 (10000 bulbs),
the 0.5499 die question, ETE family.
TRAP: report $N \times P$, not P .

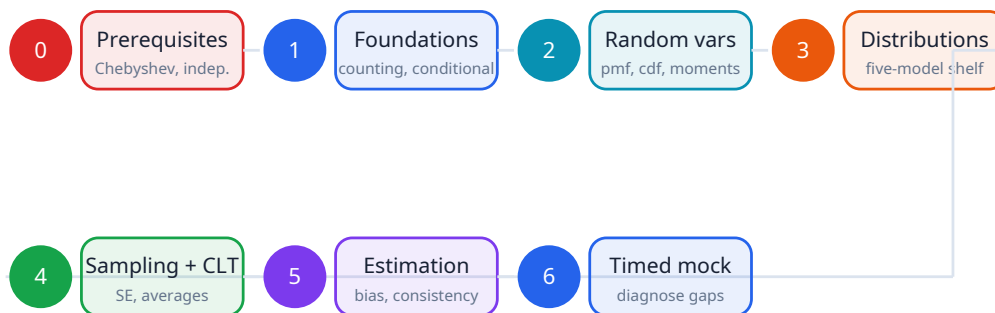
H6 GEOMETRIC $E(X) = 1/p$ (counts trials to first success)
graded: A1 short Q4 ($E = 2$).

H7 HYPERGEOMETRIC MEAN $= nK/N$ (draws WITHOUT replacement)
graded: A1 long Q3 ($4 \times 5/25 = 0.80$ exactly, both cases)
note it EQUALS the binomial mean np at the same p .

H8 THE TWO NORMAL TABLE CONVENTIONS $\phi(z)$ vs $F(z)$
 $\phi(z)$ = area from 0 to z ; $F(z)$ = cumulative from $-\infty$
differ by 0.5 for $z > 0$. DECIDE FIRST.

The hidden-layer study order and cost

Study order with dependency gates



advance only after the gate for the current pass is met

Study order and dependency gates

priority	item	why	time
1	H2	highest mark count (9+), gates nothing but is itself a whole topic	30m
2	H1	gates 4 downstream topics	25m
3	H8	every normal/CLT numeric	15m
4	H3	2 graded uses, one line to state	10m
5	H5	graded repeatedly, one idea	10m
6	H6	1 graded use	5m
7	H7	1 graded use	5m
8	H4	1 graded use, pure recall	5m

TOTAL $\approx$ 1h45

WHY THIS BLOCK IS THE BEST-SPENT TIME BEFORE THE PAPER:
it is the part of the syllabus that the slides UNDER-TAUGHT
but the papers KEEP ASKING. Cheapest marks available.